

Helios Compute Control - technical appendix

SYNTHETIC - NOT INVESTMENT ADVICE

This appendix retains reproducibility records, raw identifiers, mappings, and formula definitions. It is not the investment-committee front page.

Content identity

- Manifest SHA-256: `ad7d829e3247f537f0a9fb463e9dfa49091504abac3a94ce9fd69abdb39aeab9`
- Analysis SHA-256: `779c96335443886fa3ef2148f31a842f1eac61b985d77850b99e033ab103e62`
- Decision SHA-256: `26deaf964f55bfc190928246d99983cd3bf02b5610aec5e72c523266213ac95`
- Packet SHA-256: `7d2a2db175338697f9d6f509a56652646f06bd01362643342459fdc544d797f`

Analysis receipts

ANALYSIS	CLASSIFICATION	STATE	SPECIFICATION	RECEIPT
HX-01	ACCOUNTING_IDENTITY	REPORTED	8c611e82c119a76c9c45f8244d4960b22058ffc883c32b3d93482017cb5855c0	b9c07b32c1a6262f3b0069a79be82d445a7701b8ad77f7d094ff773b90251a32
HX-02	DESCRIPTIVE	REPORTED	4dd16a8f2fab9ac0217da45fdbb3156f53c234ee6c3b1e1143ba1a4338101858	0509303169f8bf025bc3594b4139c2f62a7cf061e639ee3eef933039118fd724
HX-03	ACCOUNTING_IDENTITY	REPORTED	3527bf827d0ab45398cc7c9811fd701083e2a5be358113a1f366f2fcc6fc332	5480f1ff78f5c6c51b38a0fce36fd36204df4e491d71e8e2d779c1b342949c5c
HX-04	DESCRIPTIVE	REPORTED	e388f757bbcddee972471756401e50cabb2389cb7b5ef90f2e555eb7dceb02a0a	0fe11718ecad1d1e6415d00cfc3efb9ea b35f2931ae10e1da69ba79cb5e326a
HX-05	PREDICTIVE_ASSOCIATION	REPORTED	869fe38c82c8037542206b8d3c5a0646792563b18f4f414a75cebe01996becb4	56ae7b892ca96d92b28b10e2894be50d26850174166ef3d4a4210a432a469216
HX-06	CAUSAL_SYNTHETIC_ONLY	REPORTED	0e1b65ba0e8dee3c2a13481cb97850a5a43ce3ab6ac091b2678138c2010e6f2b	13ce7025a4507d27baea7edc84ff4f3dde9478baf64ce720e87899fa8a8f6c20
HX-07	NOT_IDENTIFIED	ABSTAIN	1d274869a7fe75ea3c635061cb3dd615d1ceb7a273b83d4917d477f76515e3cc	73fa9bef8097c9ee38a1100648c9c5f18cd6e8856ad5aedd9ba839e3836809f7
HX-08	ACCOUNTING_IDENTITY	REPORTED	a837153dd36285f132e0c74662d034ab3543b5bela4d011d069d78086fe9ca29	9fff1fd75cee1f45e1913ad9c9cf96f816b87ee795247085f49c6ef6957067f
HX-09	SCENARIO	REPORTED	7780c967d835a7e2f5e001adc48968e348de043ce81d2ec84707a7f1ca315bfe	a3281b77b07ffcb6fc2ebd342bd371764b0f8185acfaa99313abfc8c568fe0b5

Evidence-to-model mappings

ANALYSIS	CREDIT CLASS	OBSERVED VALUE	MODEL TREATMENT
HX-05	SCENARIO_ONLY	\$20.90B modeled serviceable spend	Scenario calibration only; tier five remains unidentified
HX-06	VALUE_CREATION_BRIDGE	-0.0911 log-point synthetic ITT	No base-case credit; adoption and valuation remain scenario judgments
HX-01	BASE_CASE	ltm_revenue: 5695078389 cents	Bounded base-case evidence
HX-02	BASE_CASE	pooled_nrr: 135.36 percent	Bounded base-case evidence
HX-03	BASE_CASE	burn_multiple: 0.78 multiple	Bounded base-case evidence
HX-04	ZERO	inflated_opportunities: 48 count	0
HX-07	ZERO	No estimate retained	0
HX-08	BASE_CASE	first_close_new_shares: 1814223 shares	Bounded base-case evidence
HX-09	SCENARIO_ONLY	p10_moic: 0.00 multiple	Scenario calibration only

Formula register

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-headline-ownership	DIVIDE	hx-ownership	helios-MILESTONE-target-shares, helios-MILESTONE-fully-diluted-shares
vc-formula-base-bridge-exit-ev	MULTIPLY	helios-BASE-bridge-exit-enterprise-value	helios-BASE-bridge-terminal-revenue, helios-BASE-bridge-exit-multiple
vc-formula-base-bridge-exit-equity	ADD	helios-BASE-bridge-exit-equity	helios-BASE-bridge-exit-enterprise-value, helios-BASE-bridge-exit-cash
vc-formula-base-bridge-exit-cash	ADD	helios-BASE-bridge-exit-cash	helios-BASE-month-60-ending_cash_cents, helios-BASE-bridge-exit-cash-zero
vc-formula-base-minimum-cash	MIN	helios-BASE-minimum-cash	helios-BASE-month-01-ending_cash_cents, helios-BASE-month-02-ending_cash_cents, helios-BASE-month-03-ending_cash_cents; +57 more in model-appendix.json (ordered-list SHA-256 10c65ce3fbdc8c2cc717cbb53a28d45bae7c11395c4a7a5a98c1642c608c1b36)
vc-formula-base-funded-capital	SUM	helios-BASE-target-invested	helios-BASE-event-series-c-close-new-money, helios-BASE-event-series-c-tranche-new-money
vc-formula-base-ownership	DIVIDE	helios-BASE-ownership	helios-BASE-target-shares, helios-BASE-fully-diluted-shares
vc-formula-base-moic	DIVIDE	helios-BASE-gross-moic	helios-BASE-target-proceeds, helios-BASE-target-invested
vc-formula-base-dated-xirr	DATED_XIRR	helios-BASE-gross-xirr	helios-BASE-target-cash-flow-01, helios-BASE-target-cash-flow-02
vc-formula-base-target-proceeds	ADD	helios-BASE-target-proceeds	helios-BASE-waterfall-series_c-proceeds, helios-BASE-target-proceeds-zero
vc-formula-base-waterfall-conservation	SUM	helios-BASE-exit-value	helios-BASE-waterfall-common, helios-BASE-waterfall-series_a-proceeds, helios-BASE-waterfall-series_b-proceeds, helios-BASE-waterfall-series_c-proceeds, helios-BASE-waterfall-series_d-proceeds
vc-formula-milestone-bridge-exit-ev	MULTIPLY	helios-MILESTONE-bridge-exit-enterprise-value	helios-MILESTONE-bridge-terminal-revenue, helios-MILESTONE-bridge-exit-multiple
vc-formula-milestone-bridge-exit-equity	ADD	helios-MILESTONE-bridge-exit-equity	helios-MILESTONE-bridge-exit-enterprise-value, helios-MILESTONE-bridge-exit-cash
vc-formula-milestone-bridge-exit-cash	ADD	helios-MILESTONE-bridge-exit-cash	helios-MILESTONE-month-60-ending_cash_cents, helios-MILESTONE-bridge-exit-cash-zero
vc-formula-milestone-minimum-cash	MIN	helios-MILESTONE-minimum-cash	helios-MILESTONE-month-01-ending_cash_cents, helios-MILESTONE-month-02-ending_cash_cents, helios-MILESTONE-month-03-ending_cash_cents; +57 more in model-appendix.json (ordered-list SHA-256 a99f723cf3844e3b31a749dc4283338a491d6d3a3cdde22b48892da3112ce17)
vc-formula-milestone-funded-capital	SUM	helios-MILESTONE-target-invested	helios-MILESTONE-event-series-c-close-new-money, helios-MILESTONE-event-series-c-tranche-new-money
vc-formula-milestone-ownership	DIVIDE	helios-MILESTONE-ownership	helios-MILESTONE-target-shares, helios-MILESTONE-fully-diluted-shares
vc-formula-milestone-moic	DIVIDE	helios-MILESTONE-gross-moic	helios-MILESTONE-target-proceeds, helios-MILESTONE-target-invested
vc-formula-milestone-dated-xirr	DATED_XIRR	helios-MILESTONE-gross-xirr	helios-MILESTONE-target-cash-flow-01, helios-MILESTONE-target-cash-flow-02, helios-MILESTONE-target-cash-flow-03
vc-formula-milestone-01-funding	ADD	helios-MILESTONE-month-01-cash-before-operations	helios-MILESTONE-month-01-beginning_cash_cents, helios-MILESTONE-month-01-financing_cash_cents
vc-formula-milestone-01-ending	ADD	helios-MILESTONE-month-01-ending-cash-recomputed	helios-MILESTONE-month-01-cash-before-operations, helios-MILESTONE-month-01-operating_net_cash_flow_cents
vc-formula-milestone-12-funding	ADD	helios-MILESTONE-month-12-cash-before-operations	helios-MILESTONE-month-12-beginning_cash_cents, helios-MILESTONE-month-12-financing_cash_cents
vc-formula-milestone-12-ending	ADD	helios-MILESTONE-month-12-ending-cash-recomputed	helios-MILESTONE-month-12-cash-before-operations, helios-MILESTONE-month-12-operating_net_cash_flow_cents
vc-formula-milestone-24-funding	ADD	helios-MILESTONE-month-24-cash-before-operations	helios-MILESTONE-month-24-beginning_cash_cents, helios-MILESTONE-month-24-financing_cash_cents
vc-formula-milestone-24-ending	ADD	helios-MILESTONE-month-24-ending-cash-recomputed	helios-MILESTONE-month-24-cash-before-operations, helios-MILESTONE-month-24-operating_net_cash_flow_cents
vc-formula-milestone-36-funding	ADD	helios-MILESTONE-month-36-cash-before-operations	helios-MILESTONE-month-36-beginning_cash_cents, helios-MILESTONE-month-36-financing_cash_cents
vc-formula-milestone-36-ending	ADD	helios-MILESTONE-month-36-ending-cash-recomputed	helios-MILESTONE-month-36-cash-before-operations, helios-MILESTONE-month-36-operating_net_cash_flow_cents
vc-formula-milestone-60-funding	ADD	helios-MILESTONE-month-60-cash-before-operations	helios-MILESTONE-month-60-beginning_cash_cents, helios-MILESTONE-month-60-financing_cash_cents

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-milestone-60-ending	ADD	helios-MILESTONE-month-60-ending-cash-recomputed	helios-MILESTONE-month-60-cash-before-operations, helios-MILESTONE-month-60-operating_net_cash_flow_cents
vc-formula-milestone-target-proceeds	ADD	helios-MILESTONE-target-proceeds	helios-MILESTONE-waterfall-series_c-proceeds, helios-MILESTONE-target-proceeds-zero
vc-formula-milestone-waterfall-conservation	SUM	helios-MILESTONE-exit-value	helios-MILESTONE-waterfall-common, helios-MILESTONE-waterfall-series_a-proceeds, helios-MILESTONE-waterfall-series_b-proceeds, helios-MILESTONE-waterfall-series_c-proceeds
vc-formula-downside-bridge-exit-ev	MULTIPLY	helios-DOWNSIDE-bridge-exit-enterprise-value	helios-DOWNSIDE-bridge-terminal-revenue, helios-DOWNSIDE-bridge-exit-multiple
vc-formula-downside-bridge-exit-equity	ADD	helios-DOWNSIDE-bridge-exit-equity	helios-DOWNSIDE-bridge-exit-enterprise-value, helios-DOWNSIDE-bridge-exit-cash
vc-formula-downside-bridge-exit-cash	ADD	helios-DOWNSIDE-bridge-exit-cash	helios-DOWNSIDE-month-60-ending_cash_cents, helios-DOWNSIDE-bridge-exit-cash-zero
vc-formula-downside-minimum-cash	MIN	helios-DOWNSIDE-minimum-cash	helios-DOWNSIDE-month-01-ending_cash_cents, helios-DOWNSIDE-month-02-ending_cash_cents, helios-DOWNSIDE-month-03-ending_cash_cents; +57 more in model-appendix.json (ordered-list SHA-256 d4d09f9e067397a980660222cb6b8c58797ddd65183199349174708e1828e70f)
vc-formula-downside-funded-capital	SUM	helios-DOWNSIDE-target-invested	helios-DOWNSIDE-event-series-c-close-new-money, helios-DOWNSIDE-event-series-c-tranche-new-money
vc-formula-downside-ownership	DIVIDE	helios-DOWNSIDE-ownership	helios-DOWNSIDE-target-shares, helios-DOWNSIDE-fully-diluted-shares
vc-formula-downside-moic	DIVIDE	helios-DOWNSIDE-gross-moic	helios-DOWNSIDE-target-proceeds, helios-DOWNSIDE-target-invested
vc-formula-downside-dated-xirr	DATED_XIRR	helios-DOWNSIDE-gross-xirr	helios-DOWNSIDE-target-cash-flow-01, helios-DOWNSIDE-target-cash-flow-02
vc-formula-downside-target-proceeds	ADD	helios-DOWNSIDE-target-proceeds	helios-DOWNSIDE-waterfall-series_c-proceeds, helios-DOWNSIDE-target-proceeds-zero
vc-formula-downside-waterfall-conservation	SUM	helios-DOWNSIDE-exit-value	helios-DOWNSIDE-waterfall-common, helios-DOWNSIDE-waterfall-series_a-proceeds, helios-DOWNSIDE-waterfall-series_b-proceeds, helios-DOWNSIDE-waterfall-series_c-proceeds, helios-DOWNSIDE-waterfall-series_d-proceeds
vc-formula-financing_shortfall-bridge-exit-ev	MULTIPLY	helios-FINANCING_SHORTFALL-bridge-exit-enterprise-value	helios-FINANCING_SHORTFALL-bridge-terminal-revenue, helios-FINANCING_SHORTFALL-bridge-exit-multiple
vc-formula-financing_shortfall-bridge-exit-equity	ADD	helios-FINANCING_SHORTFALL-bridge-exit-equity	helios-FINANCING_SHORTFALL-bridge-exit-enterprise-value, helios-FINANCING_SHORTFALL-bridge-exit-cash
vc-formula-financing_shortfall-bridge-exit-cash	ADD	helios-FINANCING_SHORTFALL-bridge-exit-cash	helios-FINANCING_SHORTFALL-month-60-ending_cash_cents, helios-FINANCING_SHORTFALL-bridge-exit-cash-zero
vc-formula-financing_shortfall-minimum-cash	MIN	helios-FINANCING_SHORTFALL-minimum-cash	helios-FINANCING_SHORTFALL-month-01-ending_cash_cents, helios-FINANCING_SHORTFALL-month-02-ending_cash_cents, helios-FINANCING_SHORTFALL-month-03-ending_cash_cents; +57 more in model-appendix.json (ordered-list SHA-256 e633a4835ald7e7b9bfff93a0f30d81076e6b835c529a087137e07b29b16f6db4)
vc-formula-financing_shortfall-funded-capital	SUM	helios-FINANCING_SHORTFALL-target-invested	helios-FINANCING_SHORTFALL-event-series-c-close-new-money, helios-FINANCING_SHORTFALL-event-series-c-tranche-new-money
vc-formula-financing_shortfall-ownership	DIVIDE	helios-FINANCING_SHORTFALL-ownership	helios-FINANCING_SHORTFALL-target-shares, helios-FINANCING_SHORTFALL-fully-diluted-shares
vc-formula-financing_shortfall-moic	DIVIDE	helios-FINANCING_SHORTFALL-gross-moic	helios-FINANCING_SHORTFALL-target-proceeds, helios-FINANCING_SHORTFALL-target-invested
vc-formula-financing_shortfall-dated-xirr	DATED_XIRR	helios-FINANCING_SHORTFALL-gross-xirr	helios-FINANCING_SHORTFALL-target-cash-flow-01, helios-FINANCING_SHORTFALL-target-cash-flow-02
vc-formula-financing_shortfall-target-proceeds	ADD	helios-FINANCING_SHORTFALL-target-proceeds	helios-FINANCING_SHORTFALL-waterfall-series_c-proceeds, helios-FINANCING_SHORTFALL-target-proceeds-zero
vc-formula-financing_shortfall-waterfall-conservation	SUM	helios-FINANCING_SHORTFALL-exit-value	helios-FINANCING_SHORTFALL-waterfall-common, helios-FINANCING_SHORTFALL-waterfall-series_a-proceeds, helios-FINANCING_SHORTFALL-waterfall-series_b-proceeds, helios-FINANCING_SHORTFALL-waterfall-series_bridge-proceeds, helios-FINANCING_SHORTFALL-waterfall-series_c-proceeds
vc-formula-distribution-p10-moic	QUANTILE_P10	helios-distribution-moic-0	helios-distribution-draw-count, helios-distribution-p10-rank-index, helios-distribution-pa-th-0000-moic; +999 more in model-appendix.json (ordered-list SHA-256 39805f22ed2f5e866b67db01bb1beaab83a0731e45be30f68dec52ba8a78117a)
vc-formula-distribution-p50-moic	QUANTILE_P50	helios-distribution-moic-1	helios-distribution-draw-count, helios-distribution-p50-rank-index, helios-distribution-pa-th-0000-moic; +999 more in model-appendix.json (ordered-list SHA-256 6ad63cf6affc72f01b8004bec20070aa27be997455204ba004994204919dcdbb)

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-distribution-p90-moic	QUANTILE_P90	helios-distribution-moic-2	helios-distribution-draw-count, helios-distribution-p90-rank-index, helios-distribution-path-0000-moic; +999 more in model-appendix.json (ordered-list SHA-256 8d79186813a9d4747f00b1e5943ea091ac1e21b5c08ef95ac1e9b898c3d9465c)
vc-formula-distribution-p10-xirr	QUANTILE_P10	helios-distribution-xirr-0	helios-distribution-draw-count, helios-distribution-p10-rank-index, helios-distribution-path-0000-xirr; +999 more in model-appendix.json (ordered-list SHA-256 da8d62f556120bff66818a7e395ae71fbl6d4992fadb18578de078a7b5d4bb8)
vc-formula-distribution-p50-xirr	QUANTILE_P50	helios-distribution-xirr-1	helios-distribution-draw-count, helios-distribution-p50-rank-index, helios-distribution-path-0000-xirr; +999 more in model-appendix.json (ordered-list SHA-256 87161fe1577f84a558f170711ad1fa86c7939c6388e24c0ffe6f688d48b53aa)
vc-formula-distribution-p90-xirr	QUANTILE_P90	helios-distribution-xirr-2	helios-distribution-draw-count, helios-distribution-p90-rank-index, helios-distribution-path-0000-xirr; +999 more in model-appendix.json (ordered-list SHA-256 5c214db2af3223f647e8332e3687d52ebe7cf31f176770f24e72783108d21660)
vc-formula-distribution-probability-below-one	PROBABILITY_BELOW_ONE_PERCENT	helios-hx-09-probability_below_lx	helios-distribution-path-0000-moic, helios-distribution-path-0001-moic, helios-distribution-path-0002-moic; +997 more in model-appendix.json (ordered-list SHA-256 464817cdeae8b1f9d141d174fd02d354b7bd3d90d3e00970ffe1e5b6d3186626)
vc-formula-vc-exit_value-1-moic	DIVIDE	helios-vc-exit_value-1-gross-moic	helios-vc-exit_value-1-proceeds, helios-vc-exit_value-1-invested
vc-formula-vc-exit_value-1-xirr	DATED_XIRR	helios-vc-exit_value-1-gross-xirr	helios-vc-exit_value-1-cash-flow-01, helios-vc-exit_value-1-cash-flow-02, helios-vc-exit_value-1-cash-flow-03
vc-formula-vc-exit_value-1-ownership	DIVIDE	helios-vc-exit_value-1-ownership	helios-vc-exit_value-1-target-shares, helios-vc-exit_value-1-fully-diluted-shares
vc-formula-vc-exit_value-1-minimum-cash	MIN	helios-vc-exit_value-1-minimum-cash	helios-vc-exit_value-1-ending-cash-01, helios-vc-exit_value-1-ending-cash-02, helios-vc-exit_value-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 986078e491efef4df9c83b1e5eb6f4c00f2fd4e646e3fdb8d1065f5c477b5ac4)
vc-formula-vc-exit_value-2-moic	DIVIDE	helios-vc-exit_value-2-gross-moic	helios-vc-exit_value-2-proceeds, helios-vc-exit_value-2-invested
vc-formula-vc-exit_value-2-xirr	DATED_XIRR	helios-vc-exit_value-2-gross-xirr	helios-vc-exit_value-2-cash-flow-01, helios-vc-exit_value-2-cash-flow-02, helios-vc-exit_value-2-cash-flow-03
vc-formula-vc-exit_value-2-ownership	DIVIDE	helios-vc-exit_value-2-ownership	helios-vc-exit_value-2-target-shares, helios-vc-exit_value-2-fully-diluted-shares
vc-formula-vc-exit_value-2-minimum-cash	MIN	helios-vc-exit_value-2-minimum-cash	helios-vc-exit_value-2-ending-cash-01, helios-vc-exit_value-2-ending-cash-02, helios-vc-exit_value-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 2c29edd029d943ed81e94f3553606fd523069dc2af9c6a5dd4c2d1399f7ae75a)
vc-formula-vc-exit_value-3-moic	DIVIDE	helios-vc-exit_value-3-gross-moic	helios-vc-exit_value-3-proceeds, helios-vc-exit_value-3-invested
vc-formula-vc-exit_value-3-xirr	DATED_XIRR	helios-vc-exit_value-3-gross-xirr	helios-vc-exit_value-3-cash-flow-01, helios-vc-exit_value-3-cash-flow-02, helios-vc-exit_value-3-cash-flow-03
vc-formula-vc-exit_value-3-ownership	DIVIDE	helios-vc-exit_value-3-ownership	helios-vc-exit_value-3-target-shares, helios-vc-exit_value-3-fully-diluted-shares
vc-formula-vc-exit_value-3-minimum-cash	MIN	helios-vc-exit_value-3-minimum-cash	helios-vc-exit_value-3-ending-cash-01, helios-vc-exit_value-3-ending-cash-02, helios-vc-exit_value-3-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 01b2f439cf5c1a33fe35fa5125e82193b2af1cd5fe3dc64ee88ccbdb40424892)
vc-formula-vc-exit_date-1-moic	DIVIDE	helios-vc-exit_date-1-gross-moic	helios-vc-exit_date-1-proceeds, helios-vc-exit_date-1-invested
vc-formula-vc-exit_date-1-xirr	DATED_XIRR	helios-vc-exit_date-1-gross-xirr	helios-vc-exit_date-1-cash-flow-01, helios-vc-exit_date-1-cash-flow-02, helios-vc-exit_date-1-cash-flow-03
vc-formula-vc-exit_date-1-ownership	DIVIDE	helios-vc-exit_date-1-ownership	helios-vc-exit_date-1-target-shares, helios-vc-exit_date-1-fully-diluted-shares
vc-formula-vc-exit_date-1-minimum-cash	MIN	helios-vc-exit_date-1-minimum-cash	helios-vc-exit_date-1-ending-cash-01, helios-vc-exit_date-1-ending-cash-02, helios-vc-exit_date-1-ending-cash-03; +45 more in model-appendix.json (ordered-list SHA-256 3d01454ae3c5a08751d08de7f119670aa3e501e25addace4404ffdf7907dcd06)
vc-formula-vc-exit_date-2-moic	DIVIDE	helios-vc-exit_date-2-gross-moic	helios-vc-exit_date-2-proceeds, helios-vc-exit_date-2-invested
vc-formula-vc-exit_date-2-xirr	DATED_XIRR	helios-vc-exit_date-2-gross-xirr	helios-vc-exit_date-2-cash-flow-01, helios-vc-exit_date-2-cash-flow-02, helios-vc-exit_date-2-cash-flow-03
vc-formula-vc-exit_date-2-ownership	DIVIDE	helios-vc-exit_date-2-ownership	helios-vc-exit_date-2-target-shares, helios-vc-exit_date-2-fully-diluted-shares

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-vc-exit_date-2-minimum-cash	MIN	helios-vc-exit_date-2-minimum-cash	helios-vc-exit_date-2-ending-cash-01, helios-vc-exit_date-2-ending-cash-02, helios-vc-exit_date-2-ending-cash-03; +51 more in model-appendix.json (ordered-list SHA-256 42f309b92f99a969252f0e83cc0dc68ff37bd702438a8d0a913d532f467275)
vc-formula-vc-exit_date-3-moic	DIVIDE	helios-vc-exit_date-3-gross-moic	helios-vc-exit_date-3-proceeds, helios-vc-exit_date-3-invested
vc-formula-vc-exit_date-3-xirr	DATED_XIRR	helios-vc-exit_date-3-gross-xirr	helios-vc-exit_date-3-cash-flow-01, helios-vc-exit_date-3-cash-flow-02, helios-vc-exit_date-3-cash-flow-03
vc-formula-vc-exit_date-3-ownership	DIVIDE	helios-vc-exit_date-3-ownership	helios-vc-exit_date-3-target-shares, helios-vc-exit_date-3-fully-diluted-shares
vc-formula-vc-exit_date-3-minimum-cash	MIN	helios-vc-exit_date-3-minimum-cash	helios-vc-exit_date-3-ending-cash-01, helios-vc-exit_date-3-ending-cash-02, helios-vc-exit_date-3-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 6b0944adbac9f7650f58f4e4b1a194e1f8e9f4bb594acf0b75019069dc0df3b)
vc-formula-vc-later_round_price-1-moic	DIVIDE	helios-vc-later_round_price-1-gross-moic	helios-vc-later_round_price-1-proceeds, helios-vc-later_round_price-1-invested
vc-formula-vc-later_round_price-1-xirr	DATED_XIRR	helios-vc-later_round_price-1-gross-xirr	helios-vc-later_round_price-1-cash-flow-01, helios-vc-later_round_price-1-cash-flow-02, helios-vc-later_round_price-1-cash-flow-03
vc-formula-vc-later_round_price-1-ownership	DIVIDE	helios-vc-later_round_price-1-ownership	helios-vc-later_round_price-1-target-shares, helios-vc-later_round_price-1-fully-diluted-shares
vc-formula-vc-later_round_price-1-minimum-cash	MIN	helios-vc-later_round_price-1-minimum-cash	helios-vc-later_round_price-1-ending-cash-01, helios-vc-later_round_price-1-ending-cash-02, helios-vc-later_round_price-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 94289336e5c2d10253d82ab0dfdf4c2ead1c29e4398ea4b3faa37a13b98c4d99)
vc-formula-vc-later_round_price-2-moic	DIVIDE	helios-vc-later_round_price-2-gross-moic	helios-vc-later_round_price-2-proceeds, helios-vc-later_round_price-2-invested
vc-formula-vc-later_round_price-2-xirr	DATED_XIRR	helios-vc-later_round_price-2-gross-xirr	helios-vc-later_round_price-2-cash-flow-01, helios-vc-later_round_price-2-cash-flow-02, helios-vc-later_round_price-2-cash-flow-03
vc-formula-vc-later_round_price-2-ownership	DIVIDE	helios-vc-later_round_price-2-ownership	helios-vc-later_round_price-2-target-shares, helios-vc-later_round_price-2-fully-diluted-shares
vc-formula-vc-later_round_price-2-minimum-cash	MIN	helios-vc-later_round_price-2-minimum-cash	helios-vc-later_round_price-2-ending-cash-01, helios-vc-later_round_price-2-ending-cash-02, helios-vc-later_round_price-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 dd3148f0eb67012e754a709f2eee08d785957ef073a88f838cladec693e76f7a)
vc-formula-vc-later_round_price-3-moic	DIVIDE	helios-vc-later_round_price-3-gross-moic	helios-vc-later_round_price-3-proceeds, helios-vc-later_round_price-3-invested
vc-formula-vc-later_round_price-3-xirr	DATED_XIRR	helios-vc-later_round_price-3-gross-xirr	helios-vc-later_round_price-3-cash-flow-01, helios-vc-later_round_price-3-cash-flow-02, helios-vc-later_round_price-3-cash-flow-03
vc-formula-vc-later_round_price-3-ownership	DIVIDE	helios-vc-later_round_price-3-ownership	helios-vc-later_round_price-3-target-shares, helios-vc-later_round_price-3-fully-diluted-shares
vc-formula-vc-later_round_price-3-minimum-cash	MIN	helios-vc-later_round_price-3-minimum-cash	helios-vc-later_round_price-3-ending-cash-01, helios-vc-later_round_price-3-ending-cash-02, helios-vc-later_round_price-3-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 3e80bb8504675fbc16fel2c3dec805ba21fb73cffe36d001b9ec95e273d1396)
vc-formula-vc-milestone_state-1-moic	DIVIDE	helios-vc-milestone_state-1-gross-moic	helios-vc-milestone_state-1-proceeds, helios-vc-milestone_state-1-invested
vc-formula-vc-milestone_state-1-xirr	DATED_XIRR	helios-vc-milestone_state-1-gross-xirr	helios-vc-milestone_state-1-cash-flow-01, helios-vc-milestone_state-1-cash-flow-02
vc-formula-vc-milestone_state-1-ownership	DIVIDE	helios-vc-milestone_state-1-ownership	helios-vc-milestone_state-1-target-shares, helios-vc-milestone_state-1-fully-diluted-shares
vc-formula-vc-milestone_state-1-minimum-cash	MIN	helios-vc-milestone_state-1-minimum-cash	helios-vc-milestone_state-1-ending-cash-01, helios-vc-milestone_state-1-ending-cash-02, helios-vc-milestone_state-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 bf9cfa93b6e6c926b2fe6c0cf9c4b1a4675627551420ae5a9afc483db06acfea)
vc-formula-vc-milestone_state-2-moic	DIVIDE	helios-vc-milestone_state-2-gross-moic	helios-vc-milestone_state-2-proceeds, helios-vc-milestone_state-2-invested
vc-formula-vc-milestone_state-2-xirr	DATED_XIRR	helios-vc-milestone_state-2-gross-xirr	helios-vc-milestone_state-2-cash-flow-01, helios-vc-milestone_state-2-cash-flow-02, helios-vc-milestone_state-2-cash-flow-03
vc-formula-vc-milestone_state-2-ownership	DIVIDE	helios-vc-milestone_state-2-ownership	helios-vc-milestone_state-2-target-shares, helios-vc-milestone_state-2-fully-diluted-shares

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-vc-milestone_state-2-minimum-cash	MIN	helios-vc-milestone_state-2-minimum-cash	helios-vc-milestone_state-2-ending-cash-01, helios-vc-milestone_state-2-ending-cash-02, helios-vc-milestone_state-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 0b0946538a5afbfe72d79ab464226fb7556555557adaef8bbfd0e4185969190a)
vc-formula-vc-pool_exit_treatment-1-moic	DIVIDE	helios-vc-pool_exit_treatment-1-gross-moic	helios-vc-pool_exit_treatment-1-proceeds, helios-vc-pool_exit_treatment-1-invested
vc-formula-vc-pool_exit_treatment-1-xirr	DATED_XIRR	helios-vc-pool_exit_treatment-1-gross-xirr	helios-vc-pool_exit_treatment-1-cash-flow-01, helios-vc-pool_exit_treatment-1-cash-flow-02, helios-vc-pool_exit_treatment-1-cash-flow-03
vc-formula-vc-pool_exit_treatment-1-ownership	DIVIDE	helios-vc-pool_exit_treatment-1-ownership	helios-vc-pool_exit_treatment-1-target-shares, helios-vc-pool_exit_treatment-1-fully-diluted-shares
vc-formula-vc-pool_exit_treatment-1-minimum-cash	MIN	helios-vc-pool_exit_treatment-1-minimum-cash	helios-vc-pool_exit_treatment-1-ending-cash-01, helios-vc-pool_exit_treatment-1-ending-cash-02, helios-vc-pool_exit_treatment-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 10dc287d6e1f883fld73c4a0d5c3897a72ca1681db9ac223c26a99efcc1b1c8d)
vc-formula-vc-pool_exit_treatment-2-moic	DIVIDE	helios-vc-pool_exit_treatment-2-gross-moic	helios-vc-pool_exit_treatment-2-proceeds, helios-vc-pool_exit_treatment-2-invested
vc-formula-vc-pool_exit_treatment-2-xirr	DATED_XIRR	helios-vc-pool_exit_treatment-2-gross-xirr	helios-vc-pool_exit_treatment-2-cash-flow-01, helios-vc-pool_exit_treatment-2-cash-flow-02, helios-vc-pool_exit_treatment-2-cash-flow-03
vc-formula-vc-pool_exit_treatment-2-ownership	DIVIDE	helios-vc-pool_exit_treatment-2-ownership	helios-vc-pool_exit_treatment-2-target-shares, helios-vc-pool_exit_treatment-2-fully-diluted-shares
vc-formula-vc-pool_exit_treatment-2-minimum-cash	MIN	helios-vc-pool_exit_treatment-2-minimum-cash	helios-vc-pool_exit_treatment-2-ending-cash-01, helios-vc-pool_exit_treatment-2-ending-cash-02, helios-vc-pool_exit_treatment-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 5a7cac2fe566d98ce4bf5c2a375fe098f75cd1ebfb71f4796e90ee5d109a8f11)
vc-formula-value-ordinary-expansion-minimum_cash_delta_cents	SUBTRACT	helios-value-ordinary-expansion-minimum_cash_delta_cents	helios-value-ordinary-expansion-minimum_cash_delta_cents-result, helios-value-base-minimum_cash_delta_cents
vc-formula-value-ordinary-expansion-target_proceeds_delta_cents	SUBTRACT	helios-value-ordinary-expansion-target_proceeds_delta_cents	helios-value-ordinary-expansion-target_proceeds_delta_cents-result, helios-value-base-target_proceeds_delta_cents
vc-formula-value-ordinary-expansion-gross-xirr-delta	SUBTRACT	helios-value-ordinary-expansion-gross-xirr-delta	helios-value-ordinary-expansion-gross-xirr-delta-result, helios-value-base-gross-xirr-delta
vc-formula-value-ordinary-expansion-gross-moic-delta	SUBTRACT	helios-value-ordinary-expansion-gross-moic-delta	helios-value-ordinary-expansion-gross-moic-delta-result, helios-value-base-gross-moic-delta
vc-formula-value-optimizer-unit-economics-minimum_cash_delta_cents	SUBTRACT	helios-value-optimizer-unit-economics-minimum_cash_delta_cents	helios-value-optimizer-unit-economics-minimum_cash_delta_cents-result, helios-value-base-minimum_cash_delta_cents
vc-formula-value-optimizer-unit-economics-target_proceeds_delta_cents	SUBTRACT	helios-value-optimizer-unit-economics-target_proceeds_delta_cents	helios-value-optimizer-unit-economics-target_proceeds_delta_cents-result, helios-value-base-target_proceeds_delta_cents
vc-formula-value-optimizer-unit-economics-gross-xirr-delta	SUBTRACT	helios-value-optimizer-unit-economics-gross-xirr-delta	helios-value-optimizer-unit-economics-gross-xirr-delta-result, helios-value-base-gross-xirr-delta
vc-formula-value-optimizer-unit-economics-gross-moic-delta	SUBTRACT	helios-value-optimizer-unit-economics-gross-moic-delta	helios-value-optimizer-unit-economics-gross-moic-delta-result, helios-value-base-gross-moic-delta
vc-formula-value-sales-governance-minimum_cash_delta_cents	SUBTRACT	helios-value-sales-governance-minimum_cash_delta_cents	helios-value-sales-governance-minimum_cash_delta_cents-result, helios-value-base-minimum_cash_delta_cents
vc-formula-value-sales-governance-target_proceeds_delta_cents	SUBTRACT	helios-value-sales-governance-target_proceeds_delta_cents	helios-value-sales-governance-target_proceeds_delta_cents-result, helios-value-base-target_proceeds_delta_cents
vc-formula-value-sales-governance-gross-xirr-delta	SUBTRACT	helios-value-sales-governance-gross-xirr-delta	helios-value-sales-governance-gross-xirr-delta-result, helios-value-base-gross-xirr-delta
vc-formula-value-sales-governance-gross-moic-delta	SUBTRACT	helios-value-sales-governance-gross-moic-delta	helios-value-sales-governance-gross-moic-delta-result, helios-value-base-gross-moic-delta
vc-formula-value-combined-minimum_cash_delta_cents	SUBTRACT	helios-value-combined-minimum_cash_delta_cents	helios-value-combined-minimum_cash_delta_cents-result, helios-value-base-minimum_cash_delta_cents
vc-formula-value-combined-target_proceeds_delta_cents	SUBTRACT	helios-value-combined-target_proceeds_delta_cents	helios-value-combined-target_proceeds_delta_cents-result, helios-value-base-target_proceeds_delta_cents
vc-formula-value-combined-gross-xirr-delta	SUBTRACT	helios-value-combined-gross-xirr-delta	helios-value-combined-gross-xirr-delta-result, helios-value-base-gross-xirr-delta
vc-formula-value-combined-gross-moic-delta	SUBTRACT	helios-value-combined-gross-moic-delta	helios-value-combined-gross-moic-delta-result, helios-value-base-gross-moic-delta

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-value-standalone-proceeds-sum	SUM	helios-value-sum_standalone_target_proceeds_delta_cents	helios-value-ordinary-expansion-target_proceeds_delta_cents, helios-value-optimizer-unit-economics-target_proceeds_delta_cents, helios-value-sales-governance-target_proceeds_delta_cents
vc-formula-value-interaction	SUBTRACT	helios-value-interaction_residual_cents	helios-value-combined_target_proceeds_delta_cents, helios-value-sum_standalone_target_proceeds_delta_cents

Reproducibility boundary

All data and results are synthetic. Exact arithmetic, source locators, scenario receipts, and deterministic generation establish internal reproducibility only. They do not establish live-deal accuracy, investment approval, or real-world performance.